

NPLify · P0 Technical Baseline

Ledger Posting Design & Chart of Accounts · Draft v1.2

New XP Technologies Limited

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Status: Built against ERD & Data Model Draft v5.6 (decisions D1–D45) and NPL's answers of 10 October on accrual timing, loss events, the reseller project and variance accounts. Every worked example was produced by a small posting engine that refuses any transaction whose debits and credits do not match per currency and per owner, so the figures add up by construction. Supersedes Draft v1.1 (ERD v5.2).

Audience: NPL Finance and Management, and the New XP engineering team. Written in plain language; the account codes and field names are the ones in the ERD, with NPL's vocabulary (settlement, partner entity, markup share, sender pays / receiver pays, fee on market / fee in rate, agent rate).

What changed since v1.1. The chart of accounts grows with the v5.3–v5.6 purposes (EARN_REBATE, SHARE_RECEIVABLE, EXP_ROUNDING, EXP_LOSS) and with the purposes this draft proposes (Section 22): four variance accounts in place of one, loss receivables and the unfunded make-good pair, the reseller revenue and cost accounts, and a clearing account for company balances settled in another currency. Rebates and markup shares are accrued at conversion. A loss is booked to NPL in full and recovered share by share. The NPL-GR project keeps a full reseller P&L inside NPLify. New worked patterns: same-currency pass-through, round-up rounding, bridge-wallet hop, party-retains share with rebate, cutoff and honoured-rate variances, loss event with recoveries, invoice instalments.

1 · What the ledger is for, in one paragraph

NPL coordinates other people's money and, in one project, trades on its own account. The ledger answers, at any moment and for any past moment, three questions: **where is the money** (at which partner, in transit, in one of NPL's wallets), **whose money is it** (the client's, a sender's, or NPL's), and **what did each movement cost or earn** (NPL's margin and shares of it, rebates, partner cost, bank and network fees, rounding, losses, and the four kinds of variance). It does this with ordinary double-entry bookkeeping: every movement is a set of debits and credits that balance, nothing is ever edited or deleted, and every balance anyone sees is a sum over those entries.

2 · Rules that every posting follows

1. **One transaction per event.** A collection, a conversion, a balance conversion, a settlement, a settlement return, a confirmation, a bank-fee event, a reroute, a markup-share or rebate accrual or settlement, a loss event or recovery, and a manual adjustment each produce exactly one LEDGER_TRANSACTION that points at its source record (source_type, source_id).
2. **Balanced per currency and per owner.** Within one transaction the debits equal the credits in each currency separately, and separately again for client-tagged and company-tagged lines. A conversion

therefore has an in-currency part and an out-currency part that each balance on their own; no exchange-rate plug is ever needed, and nothing silently moves between the client's books and NPL's.

3. **Ownership is a tag on every line** (`ownership_tag` client / company) and the account code says the same (CL. / CO.). Client money becomes NPL's at one event only, margin recognition at conversion (invariant 16); company money becomes client money only where NPL absorbs a fee, tops up a shortfall or makes good a loss.
4. **A cost component on every line** (`cost_component`): principal, earnings, rebate, `partner_cost`, `bank_fee`, `network_fee`, variance, share, rounding, `fx_timing`, `cutoff_timing`, loss, and **resale** for the reseller project. Reports of margin, cost and variance are sums over this tag; no figure is stored.
5. **Accruals post with the conversion.** NPL's margin, each markup-share party's part and the partner rebate expected on the leg are all recognised at conversion, as separate transactions that name the same deal; the monthly statement settles them and any difference is a true-up, never a rewrite.
6. **Variances are booked by kind**, to four accounts: `VAR_CONVERSION` (the partner did something other than expected), `VAR_CUTOFF` (D29), `VAR_RATE_HONOUR` (D45), `VAR_FX_TIMING` (D24 and the own desk). Each can be a gain or a loss; each line names the leg or line it arose on.
7. **A loss is NPL's first.** A `LOSS_EVENT` posts the whole amount to `EXP_LOSS` and makes the client whole in the same transaction; each share the partner, the sender or the client agrees to bear is a separate, later recovery posting.
8. **A reseller project is company money throughout.** In NPL-GR the customer's price is revenue when collected, the vendor's invoice is cost as it is discharged, and the where-is-it accounts are kept under owner CO with the own desk as holder.
9. **Append-only.** A correction is a reversing transaction that names what it reverses (`reversal_of_id`) followed by the correct posting (invariant 2).
10. **Amounts stay in their own currency.** A converted total (volumes, earnings, balances, custody in the reporting currency) is a `REPORTING_VALUE`: computed at read time from the latest platform-pair market rate, carrying that rate and its time, flagged when stale, and never posted, quoted or settled (invariant 38).
11. **Rounding has an owner.** A dropped fraction stays client money in the balance; an amount added by rounding up is NPL's cost in `EXP_ROUNDING` (invariant 24).

3 · Chart of accounts

Account codes read `OWNER.HOLDER.CURRENCY.PURPOSE`. The owner is CL for client funds or CO for company. The holder says where the money is or whom it concerns: a partner code such as ALI, JETON or AQN (a `PARTNER_CONFIG`, so always within one project), `OWNDESK` for the own-desk configuration of the reseller project, `OWN` for NPL's own wallets, a receiver, a sender, a markup-share party, `PROJECT` for the project as a whole, or `NONE` for income, expense and clearing accounts. One account exists per combination actually used; accounts are created on first posting (`LEDGER_ACCOUNT`: code, ownership, holder_type, holder_id, currency, purpose, kind). Purposes marked **v1.2** are proposed by this draft and listed in Section 22.

Code	Kind	Since	What the balance means
CL.<PARTNER>.<CCY>.COLLECTED	asset	v5.2	Client money received at this partner and not yet converted (in the collection currency). In a reseller project the same purpose exists under CO.
CL.OWN.<CCY>.HELD	asset	v5.2	Client money in one of NPL's own wallets: an approved reroute, or a collection bridge waiting to be forwarded (D40). Zero otherwise (invariant 3).

Code	Kind	Since	What the balance means
CL.<PARTNER>.<CCY>.DUE	asset	v5.2	Converted client money resting at this partner, waiting to be settled: one account per project × partner × currency, the PROJECT_BALANCE.
CL.<PARTNER> / OWN.<CCY>.INTRANSIT	asset	v5.2	Client money released by this partner (or by a bridge wallet on a hop) and not yet confirmed by the receiver.
CL.<RECEIVER>.<CCY>.SHORTFALL	asset	v5.2	Client money delivered short to a receiver and still to be made good at the next settlement.
CL.PROJECT.<CCY>.PAYABLE	liability	v5.2	What NPL must deliver to the project's receivers, in total per currency. The split by receiver group is the GROUP_ENTITLEMENT view, not an account.
CL.<SENDER>.<CCY>.CREDIT	liability	v5.2	Owed back to a sender: an overpayment waiting to be refunded or applied to the next deal (SENDER_CREDIT).
CL.PROJECT.<CCY>.MAKEGOOD	asset	v1.2	Client money NPL has undertaken to replace after a loss but has not yet placed at a partner. Appears only when NPL's pool at the partner cannot fund the make-good at once; cleared when NPL's funds land (Pattern L).
CO.<PARTNER>.<CCY>.POOL	asset	v5.2	NPL's own money resting at this partner: margin captured at conversion (EARNINGS_RECEIVABLE), plus anything NPL transferred there. Settled on the partner's fee cycle or netted on a reroute.
CO.OWN.<CCY>.WALLET	asset	v5.2	NPL's own money in its own wallets: recovered dues, rebates and shares received, gas reserves.
CO.<PARTNER>.<CCY>.REBATE_RECEIVABLE	asset	v5.2	Rebate a partner owes NPL on NPLify deals, accrued at conversion and reconciled against the partner's monthly statement (D27).
CO.<PARTY>.<CCY>.SHARE_RECEIVABLE	asset	v5.3	NPL's remainder of the markup owed by a party that holds the converted funds (settlement_mode party_retains, D26); reconciled monthly.
CO.<PARTY>.<CCY>.SHARE_PAYABLE	liability	v5.2	Markup share owed by NPL to a markup-share party (settlement_mode npl_pays, D26).
CO.<PARTNER> / <SENDER>.<CCY>.LOSS_RECEIVABLE	asset	v1.2	The partner's or the sender's agreed share of a loss event, owed to NPL after NPL booked the whole loss first (Pattern L).
CO.NONE.<CCY>.LOSS_PAYABLE	liability	v1.2	The unfunded part of a make-good: what NPL still has to put in after a loss. Mirrors CL.PROJECT.<CCY>.MAKEGOOD and is cleared by the same transfer.
CO.OWNDESK.<CCY>.COLLECTED / DUE / INTRANSIT	asset	v1.2	In a reseller project (NPL-GR) the money is NPL's from the moment it arrives, so the where-is-it purposes are used under owner CO with the own desk as holder (Pattern M).
CO.NONE.<CCY>.EARN_GROSS	income	v5.2	Margin recognised at conversion (sender and receiver fee parts), before partner cost. Markup shares are debited here, so the balance is NPL's margin after shares.
CO.NONE.<CCY>.EARN_REBATE	income	v5.3	Rebate income, kept apart from EARN_GROSS and never in a markup-share base (D27).
CO.NONE.<CCY>.RESELL_REVENUE	income	v1.2	The retail price a reseller customer pays NPL-GR, recognised when collected, in the currency collected (Pattern M).

Code	Kind	Since	What the balance means
CO.NONE.<CCY>.RESELL_COST	expense	v1.2	The vendor's invoice, recognised as each payout line discharges part of it, in the currency paid, at the rate the deal was priced on (Pattern M).
CO.NONE.<CCY>.EXP_PARTNER	expense	v5.2	Partner cost where the partner states it separately (fee_on_market). Zero by construction for fee_in_rate partners and the own desk.
CO.NONE.<CCY>.EXP_BANKFEE	expense	v5.2	Bank and payout fees NPL absorbs: gross-ups, shortfall top-ups, bounce fees, covered fees (D41).
CO.NONE.<CCY>.EXP_NETWORK	expense	v5.2	Crypto network fees NPL absorbs: a refund, a bridge hop, a forwarded reroute.
CO.NONE.<CCY>.EXP_ROUNDING	expense	v5.3	The amount added when a payout is rounded up to the pair's unit (D25).
CO.NONE.<CCY>.EXP_LOSS	expense	v5.3	Loss events. The whole loss is booked here first; each recovered share is credited back as it is agreed (Pattern L).
CO.NONE.<CCY>.VAR_CONVERSION	expense	v1.2	Partner converted differently from the engine's expectation (partner_calculates), and basis variance on a reroute. Gain or loss. Was VARIANCE.
CO.NONE.<CCY>.VAR_CUTOFF	expense	v1.2	Rate stamped at lock versus the rate actually used when the conversion ran after the partner's cutoff; NPL absorbs it (D29).
CO.NONE.<CCY>.VAR_RATE_HONOUR	expense	v1.2	An expired quote's rate honoured by an approved override versus the partner's rate on the day of conversion; NPL's cost, chosen (D45).
CO.NONE.<CCY>.VAR_FX_TIMING	expense	v1.2	An obligation priced on one day and discharged on another: the rate difference on invoice instalments (D24) and on an own-desk conversion after the customer's price was fixed.
CO.NONE.<CCY>.FX_CLEARING	asset	v1.2	Used only when a company balance in one currency is settled in another (a EUR share receivable paid in USDT): the receivable clears in its own currency, the receipt lands in its own, and the two legs meet here. Its balances are NPL's open currency position, valued through REPORTING_VALUE.

Debit-normal accounts (assets, expenses) grow with debits; credit-normal accounts (liabilities, income) grow with credits. A variance account with a credit balance is a gain. ADVANCE does not exist because NPL never advances money to a receiver (invariant 16). The former single VARIANCE purpose is replaced by the four VAR_ accounts; the former referral payable is SHARE_PAYABLE (D37).

4 · Cost components, owners and sources

Cost component	Used on	Meaning
principal	client money moving; reseller funds moving	The amount itself: collected, converted, released, confirmed, returned.
earnings	POOL, EARN_GROSS, WALLET on dues recovery	NPL's margin (sender and receiver fee parts) and where it rests.
partner_cost	EXP_PARTNER	A fee_on_market partner's stated fee, deducted at transaction level.

Cost component	Used on	Meaning
rebate	REBATE_RECEIVABLE, EARN_REBATE, WALLET	A partner's rebate on NPLify deals: accrual, receipt, true-up.
share	EARN_GROSS (debit), SHARE_PAYABLE, SHARE_RECEIVABLE	A markup-share party's part of the net markup and its settlement.
bank_fee	EXP_BANKFEE, SHORTFALL, POOL, DUE	Bank and payout fees: waived, carried forward, absorbed, covered, bounce.
network_fee	EXP_NETWORK, WALLET, COLLECTED	Crypto network fees on refunds, hops and forwards.
rounding	DUE, EXP_ROUNDING, POOL	The signed residual of the pair's rounding option.
variance	VAR_CONVERSION, VAR_RATE_HONOUR, VAR_FX_TIMING	Expected versus actual, by kind.
cutoff_timing	VAR_CUTOFF	Rate at lock versus rate used after the partner's cutoff.
fx_timing	VAR_FX_TIMING	Obligation priced on one day, discharged on another.
loss	EXP_LOSS, LOSS_RECEIVABLE, POOL, WALLET, COLLECTED, PAYABLE	A loss event, its make-good and its recoveries.
resale	RESELL_REVENUE, RESELL_COST	The reseller project's retail price and vendor cost.

Transaction sources (LEDGER_TRANSACTION.source_type): Collection, Conversion, BalanceConversion, Settlement, SettlementReturn, Confirmation, BankFeeEvent, Reroute, MarkupShare, Rebate, LossEvent, Adjustment. Section 20 lists what each one posts; Section 22 asks for the source record a loss recovery points at.

5 · Reading a posting table

Each worked example shows one transaction as a table. *Debit* and *Credit* are the amounts; *Owner* is the ownership tag; *Cost component* is the breakdown tag; *Why* says in plain words what that line records. The running example is project Evo: Sender A, Receiver X (receiver group A, the default group) with entities Entity X1 and Entity X2; partner Ali (fee_in_rate: its cost is inside its rate), partner Jeton (fee_on_market + 0.50 %) and partner Aquanow (fee_on_market + 0.80 %, rebate 0.3 % of the amount). Rates: market 0.9000 EUR per USDT, Ali 0.8950. Fee structure: basis agent_rate, fixed, 1.00 % in total, sender pays 0.40 % and receiver pays 0.60 %. The arithmetic is in the Calculation Specification; the headline figures for deal 1 are:

Figure	Deal 1 (10,000 USDT at Ali)	How
Sender rate	0.89142	$0.895 \times (1 - 0.40 \%)$
Gross out	8,914.20 EUR	$10,000 \times \text{sender rate}$
Receiver pays	53.49 EUR	$\text{gross out} \times 0.60 \%$
Client net (entitlement)	8,860.71 EUR	$\text{gross out} - \text{receiver part}$
Converted at Ali	8,950.00 EUR	$10,000 \times 0.895$

Figure	Deal 1 (10,000 USDT at Ali)	How
NPL earnings	89.29 EUR	converted – client net (= sender part 35.80 + receiver part 53.49)
Paid out in whole euros / dust	8,860.00 / 0.71 EUR	amount_rounding truncate_unit; the dust stays client money

6 · Pattern A — direct collection, conversion and accumulation

The everyday case: the sender pays, the partner converts, the client balance at that partner grows until it is settled. Collect-first and quote-first deals post identically; only the moment pricing is fixed differs. A quote extended any number of times (D45) still posts once, at conversion.

T1 · Collection of deal 1: 10,000 USDT arrive at Ali's USDT wallet endpoint

(COLLECTION_RECEIVING_ENDPOINT, kind wallet) source: Collection

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.ALI.USDT.COLLECTED	10,000.00		USDT	client	principal	client money now sits at Ali, unconverted; the account is per partner, the endpoint only says which vehicle's wallet
CL.PROJECT.USDT.PAYABLE		10,000.00	USDT	client	principal	NPL owes this onward on the client's behalf

T2 · Conversion of deal 1 at Ali (fee_in_rate): 10,000 USDT → 8,950.00 EUR at 0.895

source: Conversion

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.PROJECT.USDT.PAYABLE	10,000.00		USDT	client	principal	the USDT obligation is discharged by the conversion
CL.ALI.USDT.COLLECTED		10,000.00	USDT	client	principal	the USDT left the collected position
CL.ALI.EUR.DUE	8,860.00		EUR	client	principal	client net in whole euros, waiting to be settled
CL.ALI.EUR.DUE	0.71		EUR	client	rounding	the cents below one euro (amount_rounding truncate_unit) stay client money
CO.ALI.EUR.POOL	89.29		EUR	company	earnings	NPL's margin, physically still at Ali (EARNINGS_RECEIVABLE)
CL.PROJECT.EUR.PAYABLE		8,860.71	EUR	client	principal	owed to the receiver group, now in EUR
CO.NONE.EUR.EARN_GROSS		89.29	EUR	company	earnings	margin recognised: the client's claim shrinks from 10,000 USDT to its EUR entitlement, the difference is NPL's

Ownership changes here and only here: 89.29 EUR becomes company money, because the client's claim is re-denominated at the sender's price while the partner delivers at its rate. Ali's own cost is inside its rate (fee_in_rate), so no partner-cost line exists; the inferred cost against market, 50.00 EUR, is kept on the deal as information only. Nothing differs for a leg split across two fee tiers (D21): the fee parts are the blended sum and the posting is identical. The customer-facing rate and fee (D22) are print-only and never post.

Balances after deal 1

Account	Ccy	Balance	Meaning
CL.ALI.USDT.COLLECTED	USDT	0.00	nothing left unconverted
CL.ALI.EUR.DUE	EUR	8,860.71	client balance at Ali (PROJECT_BALANCE)
CL.PROJECT.EUR.PAYABLE	EUR	8,860.71	owed to receivers; equals group entitlement

Account	Ccy	Balance	Meaning
CO.ALI.EUR.POOL	EUR	89.29	NPL's margin resting at Ali
CO.NONE.EUR.EARN_GROSS	EUR	89.29	margin recognised to date

T3 · Same-currency pass-through (D33): 5,000 EUR collected by bank transfer and converted at rate 1 — fee 1.00 % captured, client net 4,950.12 EUR source: Conversion

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.ALI.EUR.COLLECTED	5,000.00		EUR	client	principal	collection (shown with the conversion)
CL.PROJECT.EUR.PAYABLE		5,000.00	EUR	client	principal	collection
CL.PROJECT.EUR.PAYABLE	5,000.00		EUR	client	principal	conversion: the obligation is re-stated at the sender's price
CL.ALI.EUR.COLLECTED		5,000.00	EUR	client	principal	
CL.ALI.EUR.DUE	4,950.00		EUR	client	principal	client net, available to settle
CL.ALI.EUR.DUE	0.12		EUR	client	rounding	
CO.ALI.EUR.POOL	49.88		EUR	company	earnings	
CL.PROJECT.EUR.PAYABLE		4,950.12	EUR	client	principal	
CO.NONE.EUR.EARN_GROSS		49.88	EUR	company	earnings	the fee is earned at conversion even though no currency changes

A pair with the same currency on both sides converts at rate 1 like any other pair (invariant 32): the in-leg and the out-leg happen to be in one currency, and the balance arithmetic is unchanged.

7 · Pattern B — part collection

A deal may be collected in parts, by any method; the obligation grows with each part and the conversion waits for the last one (or for the partner's cash count).

T4 · Collection of deal 2, part 1 of 2: 4,000 USDT source: Collection

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.ALI.USDT.COLLECTED	4,000.00		USDT	client	principal	first part
CL.PROJECT.USDT.PAYABLE		4,000.00	USDT	client	principal	

T5 · Collection of deal 2, part 2 of 2: 1,000 USDT (next day) source: Collection

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.ALI.USDT.COLLECTED	1,000.00		USDT	client	principal	second part completes the deal
CL.PROJECT.USDT.PAYABLE		1,000.00	USDT	client	principal	

T6 • Conversion of deal 2 at Ali: 5,000 USDT → 4,475.00 EUR source: Conversion

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.PROJECT.USDT.PAYABLE	5,000.00		USDT	client	principal	
CL.ALI.USDT.COLLECTED		5,000.00	USDT	client	principal	
CL.ALI.EUR.DUE	4,430.00		EUR	client	principal	
CL.ALI.EUR.DUE	0.36		EUR	client	rounding	
CO.ALI.EUR.POOL	44.64		EUR	company	earnings	
CL.PROJECT.EUR.PAYABLE		4,430.36	EUR	client	principal	
CO.NONE.EUR.EARN_GROSS		44.64	EUR	company	earnings	

Parts are posted as they arrive; the conversion is one transaction once the deal is fully collected. A cash part is verified only after the partner's count (counted_at, D30) and posts then; the cash amount itself was rounded to the pair's cash unit at quote (cash_rounding, D25), so no residual arises on the collection side.

8 • Pattern C — settlement, confirmation, bank fees, shortfalls, rounding

A settlement drains a partner's holding into one or many lines; each line is confirmed by its receiver in full or short. The bank-fee treatment of the project or of a FEE_DECISION decides who bears a fee the bank took.

T7 • Settlement from Ali: 18,241.00 EUR in two lines (Entity X1 9,000.00, Entity X2 9,241.00)

source: Settlement

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.ALI.EUR.INTRANSIT	9,000.00		EUR	client	principal	line 1 released by Ali
CL.ALI.EUR.INTRANSIT	9,241.00		EUR	client	principal	line 2 released by Ali
CL.ALI.EUR.DUE		18,241.00	EUR	client	principal	whole euros leave DUE; 0.19 EUR of rounding dust stays

Settle to the residual (invariant 4): the client balance at Ali drops from 18,241.19 to 0.19 EUR. Both lines pay entities of the same receiver and each names the leg it settles in deal_id, so no offset arises. The transaction posts when the partner releases the money, which for a bank rail is no earlier than execution_due_date (approval before the partner's cutoff → same day, after it → next business day; weekends and HOLIDAY rows skipped, D29, D34). Approval and release are two moments; only the release posts.

T8 • Confirmation, line 1 full: Entity X1 received 9,000.00 EUR source: Confirmation

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.PROJECT.EUR.PAYABLE	9,000.00		EUR	client	principal	the obligation is discharged
CL.ALI.EUR.INTRANSIT		9,000.00	EUR	client	principal	the money reached the receiver

T9 • Confirmation, line 2 short: Entity X2 received 9,191.00 EUR, the bank took 50.00 (treatment carry_forward) source: Confirmation

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.PROJECT.EUR.PAYABLE	9,191.00		EUR	client	principal	delivered part
CL.RECEIVER_X.EUR.SHORTFALL	50.00		EUR	client	bank_fee	still owed to the receiver, topped up at the next settlement
CL.ALI.EUR.INTRANSIT		9,241.00	EUR	client	principal	nothing is left in transit

With treatment *waived* the second line would instead be Dr CL.PROJECT.EUR.PAYABLE 50: the client bears the fee and its claim falls by 50. With *absorbed* the fee is expected up front and funded as in T10. A fee *covered* by a FEE_DECISION (D41: Finance recommends, Management approves) posts exactly as absorbed.

T10 • Shortfall top-up funded from NPL's pool at Ali (BANK_FEE_EVENT applied to the next settlement) source: BankFeeEvent

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CO.NONE.EUR.EXP_BANKFEE	50.00		EUR	company	bank_fee	NPL bears the cost
CO.ALI.EUR.POOL		50.00	EUR	company	bank_fee	paid out of NPL's money resting at Ali
CL.ALI.EUR.DUE	50.00		EUR	client	bank_fee	the 50 EUR is client money again, in balance at Ali
CL.RECEIVER_X.EUR.SHORTFALL		50.00	EUR	client	bank_fee	the shortfall is cleared

The 50 EUR now sits in DUE and goes out with the next settlement like any other balance. Company → client transfers like this one are the only way company money becomes client money (besides a loss make-good, Pattern L).

Balances after settlement and confirmation

Account	Ccy	Balance	Meaning
CL.ALI.EUR.DUE	EUR	50.19	dust plus the topped-up 50 EUR, out with the next settlement
CL.ALI.EUR.INTRANSIT	EUR	0.00	nothing in transit
CL.PROJECT.EUR.PAYABLE	EUR	50.19	the 50 EUR still owed to Receiver X plus the dust
CL.RECEIVER_X.EUR.SHORTFALL	EUR	0.00	cleared
CO.ALI.EUR.POOL	EUR	133.81	NPL's margin at Ali after funding the top-up
CO.NONE.EUR.EXP_BANKFEE	EUR	50.00	bank fees NPL bore

T11 • Conversion with round-up (D25): 3,000 USDT → 97,500.00 THB at 32.50; client net 96,527.34, paid out as 97,000.00 (round_up_n, unit 1,000) source: Conversion

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.PROJECT.USDT.PAYABLE	3,000.00		USDT	client	principal	(collection of the 3,000 USDT assumed posted as T1)
CL.ALI.USDT.COLLECTED		3,000.00	USDT	client	principal	
CL.ALI.THB.DUE	96,527.34		THB	client	principal	client net as priced
CL.ALI.THB.DUE	472.66		THB	client	rounding	the amount added by rounding up: client money from now on
CO.ALI.THB.POOL	500.00		THB	company	earnings	NPL's margin less the round-up it funded
CO.NONE.THB.EXP_ROUNDING	472.66		THB	company	rounding	the cost of rounding up (rounding_residual is negative)
CL.PROJECT.THB.PAYABLE		97,000.00	THB	client	principal	the obligation is the rounded payout
CO.NONE.THB.EARN_GROSS		972.66	THB	company	earnings	margin as priced

The residual is signed: -472.66 THB here, so NPL funds the round-up from its margin and the client's claim is the whole 97,000. With truncate_unit (T2) the sign is positive and the dust stays client money. Every rounding difference therefore has an owner (invariant 24).

9 • Pattern D — contingency routes and the bridge wallet

NPL's own wallets hold client money in three cases only (invariant 3): an approved reroute, a collection bridge for a network the partner lacks, and the own desk. The first two are shown here; the own desk is Pattern M.

T12 • Deal 3, Ali unavailable: 10,000 USDT collected into NPL's contingency wallet under an approved REROUTE source: Collection

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.OWN.USDT.HELD	10,000.00		USDT	client	principal	client money in custody, with deal, reason and age
CL.PROJECT.USDT.PAYABLE		10,000.00	USDT	client	principal	

Only an approved REROUTE, an own-desk leg or a bridge collection may debit CL.OWN.*.HELD (invariant 3). The custody view shows this balance until it is forwarded.

T13 • Forward to Jeton with recover dues: 10,000 USDT of client money credited at Jeton, 30.00 USDT of NPL's dues netted source: Reroute

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.JETON.USDT.COLLECTED	10,000.00		USDT	client	principal	Jeton credits the full client amount
CL.OWN.USDT.HELD		10,000.00	USDT	client	principal	custody ends
CO.OWN.USDT.WALLET	30.00		USDT	company	earnings	NPL keeps 30 USDT in its wallet instead of sending it
CO.JETON.USDT.POOL		30.00	USDT	company	earnings	Jeton's debt to NPL falls by the same 30 USDT

NPL physically sends 9,970.00 USDT; Jeton treats it as 10,000 of client money because it owed NPL 30.00 USDT of earnings from earlier deals (partner_agreement_ref). The recovery is an asset swap, never income, and names the deal it recovers (invariant 15). **Nothing is withheld from the client amount:** NPL's margin is captured when Jeton converts (T14).

T14 • Conversion of deal 3 at Jeton (fee_on_market, 0.50 %): 10,000 USDT → 9,000.00 EUR at market, Jeton keeps 45.00 source: Conversion

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.PROJECT.USDT.PAYABLE	10,000.00		USDT	client	principal	
CL.JETON.USDT.COLLECTED		10,000.00	USDT	client	principal	
CL.JETON.EUR.DUE	8,910.00		EUR	client	principal	client net, whole euros
CL.JETON.EUR.DUE	0.22		EUR	client	rounding	
CO.JETON.EUR.POOL	44.78		EUR	company	earnings	NPL's net margin resting at Jeton
CO.NONE.EUR.EXP_PARTNER	45.00		EUR	company	partner_cost	Jeton's stated fee, deducted at transaction level
CL.PROJECT.EUR.PAYABLE		8,910.22	EUR	client	principal	
CO.NONE.EUR.EARN_GROSS		89.78	EUR	company	earnings	gross margin before partner cost

Total cost to the client = partner cost 45.00 + NPL earnings 44.78 = 89.78 EUR. For a fee_in_rate partner (T2) the partner cost is inside the rate and no EXP_PARTNER line exists.

Balances after the contingency route

Account	Ccy	Balance	Meaning
CL.OWN.USDT.HELD	USDT	0.00	custody ended
CL.JETON.EUR.DUE	EUR	8,910.22	client balance at Jeton
CO.JETON.EUR.POOL	EUR	44.78	NPL's net margin at Jeton
CO.OWN.USDT.WALLET	USDT	30.00	recovered dues in NPL's wallet
CO.NONE.EUR.EXP_PARTNER	EUR	45.00	Jeton's stated fees to date

T15 · Deal 4 (D40): the sender can only pay on ERC-20, which Ali does not accept; 10,000 USDT arrive on NPL's bridge wallet source: Collection

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.OWN.USDT.HELD	10,000.00		USDT	client	principal	client money on the bridge (OWN_WALLET.role collection_bridge), to be forwarded within custody_max_hours
CL.PROJECT.USDT.PAYABLE		10,000.00	USDT	client	principal	

T16 · Hop: the bridge forwards 10,000 USDT to Ali on TRC-20 (a SETTLEMENT_LINE to the partner_transit receiver); gas 0.0012 ETH paid by NPL source: Settlement

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.OWN.USDT.INTRANSIT	10,000.00		USDT	client	principal	left the bridge, not yet credited by Ali
CL.OWN.USDT.HELD		10,000.00	USDT	client	principal	custody ends when Ali credits (T17)
CO.NONE.ETH.EXP_NETWORK	0.0012		ETH	company	network_fee	network fee of the hop, NPL's cost (project network_fee_policy)
CO.OWN.ETH.WALLET		0.0012	ETH	company	network_fee	paid from NPL's gas reserve

The hop line carries deal_id (invariant 19) so the entitlement view keeps the leg's group; nothing is earned or charged to the client on a hop.

T17 · Ali credits the hop: 10,000 USDT now sit at Ali as collected client money source: Confirmation

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.ALI.USDT.COLLECTED	10,000.00		USDT	client	principal	the leg continues at its partner exactly as T1
CL.OWN.USDT.INTRANSIT		10,000.00	USDT	client	principal	

10 · Pattern E — cancellations, reversals and voids

T18 · Erroneous posting: a collection of 2,000 USDT was booked against the wrong deal (as posted) source: Collection

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.ALI.USDT.COLLECTED	2,000.00		USDT	client	principal	booked in error
CL.PROJECT.USDT.PAYABLE		2,000.00	USDT	client	principal	

T19 · Reversal of T18 (LEDGER_TRANSACTION.reversal_of_id = T18), then T18 is re-posted correctly source: Adjustment

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.PROJECT.USDT.PAYABLE	2,000.00		USDT	client	principal	mirror image of T18
CL.ALI.USDT.COLLECTED		2,000.00	USDT	client	principal	

Nothing is edited or deleted: the wrong transaction stays, the reversal points at it, and the correct transaction follows (invariant 2). Any report at any past moment still adds up.

T20 • Collection of deal 5: 3,000 USDT arrive at Ali (the deal is voided afterwards) source: Collection

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.ALI.USDT.COLLECTED	3,000.00		USDT	client	principal	
CL.PROJECT.USDT.PAYABLE		3,000.00	USDT	client	principal	

T21 • Deal 5 voided after collection: refund to the sender in USDT, network fee 2.00 USDT charged back to the sender (project_network_fee_policy) source: Adjustment

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.PROJECT.USDT.PAYABLE	3,000.00		USDT	client	principal	the obligation ends
CL.ALI.USDT.COLLECTED		2,998.00	USDT	client	principal	returned to the sender's wallet
CL.ALI.USDT.COLLECTED		2.00	USDT	client	network_fee	paid to the network; the sender bears it

If the policy, or a FEE_DECISION for this transaction, says NPL absorbs the fee, the last line becomes Dr CO.NONE.USDT.EXP_NETWORK 2.00 / Cr CO.ALI.USDT.POOL 2.00 and the sender receives the full 3,000. Management approves the disposition of any post-money void.

11 • Pattern F — rejected payout and return leg

T22 • Deal 6 collected and converted at Ali: 2,000 USDT → 1,550.00 GBP (collection and conversion shown together) source: Conversion

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.ALI.USDT.COLLECTED	2,000.00		USDT	client	principal	collection
CL.PROJECT.USDT.PAYABLE		2,000.00	USDT	client	principal	collection
CL.PROJECT.USDT.PAYABLE	2,000.00		USDT	client	principal	conversion, USDT leg
CL.ALI.USDT.COLLECTED		2,000.00	USDT	client	principal	conversion, USDT leg
CL.ALI.GBP.DUE	1,534.00		GBP	client	principal	
CL.ALI.GBP.DUE	0.54		GBP	client	rounding	
CO.ALI.GBP.POOL	15.46		GBP	company	earnings	
CL.PROJECT.GBP.PAYABLE		1,534.54	GBP	client	principal	
CO.NONE.GBP.EARN_GROSS		15.46	GBP	company	earnings	

T23 • Settlement of deal 6: 1,534.00 GBP released to the receiver's bank (the line names deal 6 in deal_id) source: Settlement

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.ALI.GBP.INTRANSIT	1,534.00		GBP	client	principal	
CL.ALI.GBP.DUE		1,534.00	GBP	client	principal	

T24 • The receiving bank rejects the transfer: 1,519.00 GBP come back to Ali, the bank kept 15.00 (SETTLEMENT_RETURN; line outcome = returned) source: SettlementReturn

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.ALI.GBP.DUE	1,519.00		GBP	client	principal	back in the client balance at Ali
CO.NONE.GBP.EXP_BANKFEE	15.00		GBP	company	bank_fee	bounce fee absorbed by NPL (treatment absorbed)
CL.ALI.GBP.INTRANSIT		1,534.00	GBP	client	principal	nothing is in transit any more
CL.ALI.GBP.DUE	15.00		GBP	client	bank_fee	NPL makes the client whole
CO.ALI.GBP.POOL		15.00	GBP	company	bank_fee	funded from NPL's pool at Ali

The failed line is reversed before anything else happens (invariant 13), so the entitlement of the group deal 6 is attributed to is restored. The money can now stay in balance for a later payout or fund a return leg (T25).

T25 • Return leg: 1,534.00 GBP converted back to 1,971.19 USDT at Ali (rate 1.285); markup waived by FEE_DECISION source: Conversion

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.PROJECT.GBP.PAYABLE	1,534.00		GBP	client	principal	GBP obligation ends
CL.ALI.GBP.DUE		1,534.00	GBP	client	principal	
CL.ALI.USDT.DUE	1,971.19		USDT	client	principal	USDT owed to the sender's return receiver
CL.PROJECT.USDT.PAYABLE		1,971.19	USDT	client	principal	

NPL's margin from T22 (15.46 GBP) stays in the pool: it was captured at the first conversion and is never at risk from a return. The round trip's rate loss falls on the client's funds. Had the FEE_DECISION applied a markup, part of the 1,971.19 USDT would have gone to CO.ALI.USDT.POOL and EARN_GROSS exactly as in T2. The return leg inherits its parent's group (D13); the payout to the sender's return receiver follows T23 and T8 with deal_id = the return leg (invariant 19).

12 • Pattern G — inter-group offsets

An offset is not an account. It arises only when a counterparty receiver of one group is paid from a leg attributed to another group (invariant 17): the settlement posts exactly as T7 and T8, and the OFFSET view, reading GROUP_ENTITLEMENT, shows one group owed and the other holding its flow. A project with a single group never has an offset; a cross-group payout needs Management approval (cross_group_payout) and tighter aging. No ledger line differs, which is the point: the money and the claim are right, only the attribution needs watching.

13 • Pattern H — balance conversion

T26 • Opening position: 5,000 SGD of converted client balance at Ali from earlier deals (summarised) source: Conversion

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.ALI.SGD.DUE	5,000.00		SGD	client	principal	balance built up by earlier SGD deals
CL.PROJECT.SGD.PAYABLE		5,000.00	SGD	client	principal	

T27 • Balance conversion at Ali: 5,000 SGD of client balance → 3,700.00 USDT at 0.74, no markup (serves Group B) source: BalanceConversion

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.PROJECT.SGD.PAYABLE	5,000.00		SGD	client	principal	SGD obligation ends
CL.ALI.SGD.DUE		5,000.00	SGD	client	principal	SGD balance leaves
CL.ALI.USDT.DUE	3,700.00		USDT	client	principal	USDT balance arrives, whole units
CL.PROJECT.USDT.PAYABLE		3,700.00	USDT	client	principal	same obligation, new currency

No sender, receiver or collection: Finance requests, Management approves. ENTITLEMENT_REATTRIBUTION moves Group B's SGD entitlement into USDT; an excess over Group B's SGD entitlement is another group's flow and is booked as an approved offset. A banking-ready balance (USD, SGD) waits for conversion_due_date like a deal leg (D29); a markup, if any, is a FEE_DECISION and posts as in T2.

14 • Pattern I — overpayment and sender credit

T28 • Overpayment: a sender sends 10,050 USDT against a 10,000 USDT deal; 50 USDT becomes sender credit source: Collection

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.ALI.USDT.COLLECTED	10,050.00		USDT	client	principal	all of it is client money at Ali
CL.PROJECT.USDT.PAYABLE		10,000.00	USDT	client	principal	the deal
CL.SENDER_A.USDT.CREDIT		50.00	USDT	client	principal	owed back to the sender (SENDER_CREDIT), or applied to the next deal

T29 • The credit is applied to the sender's next deal of 5,000 USDT: the sender pays 4,950 source: Collection

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.ALI.USDT.COLLECTED	4,950.00		USDT	client	principal	what actually arrives
CL.SENDER_A.USDT.CREDIT	50.00		USDT	client	principal	the credit is consumed (state applied)
CL.PROJECT.USDT.PAYABLE		5,000.00	USDT	client	principal	the full deal is collected

A refund instead of an application is a payout to the sender's return receiver: Dr CL.SENDER_A.USDT.CREDIT 50 / Cr CL.ALI.USDT.COLLECTED 50 through an ordinary settlement line and confirmation. Sender credits are client money and are never netted against anything NPL is owed.

15 · Pattern J — markup shares and partner rebates

Both are accrued with the conversion they arise on (NPL, 10 October) and settled monthly against a statement. A share is a part of NPL's net markup owed to a party; a rebate is NPL's own income from a partner, kept apart from the margin and never in a share base (D26, D27).

T30 · Markup share, npl_pays (D26): an external markup-share party earns 25 % of NPL's earnings on deal 3 = 11.20 EUR, accrued with the conversion source: MarkupShare

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CO.NONE.EUR.EARN_GROSS	11.20		EUR	company	share	reduces NPL's margin
CO.PARTY_P.EUR.SHARE_PAYABLE		11.20	EUR	company	share	owed to the party (MARKUP_SHARE_ACCRUAL, direction payable)

Basis share_of_earnings. A fixed_pct party is priced into the sender rate instead and the posting has the same shape with the amount from the Calculation Specification. The accrual is posted in the same moment as T14, as its own transaction with source MarkupShare.

T31 · Month-end: NPL pays the party 11.20 EUR from its own wallet source: MarkupShare

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CO.PARTY_P.EUR.SHARE_PAYABLE	11.20		EUR	company	share	accrual state settled, paid_at set
CO.OWN.EUR.WALLET		11.20	EUR	company	share	or from the pool at a partner, when the partner pays the party on NPL's instruction

T32 · Raeen sub-account at Aquanow (party_retains, D26): 10,000 USDT → 8,928.00 EUR; sender pays 4 % on market, Aquanow charges 0.80 %; net markup 288.00 EUR shared 50 / 50

source: Conversion

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.PROJECT.USDT.PAYABLE	10,000.00		USDT	client	principal	
CL.AQN.USDT.COLLECTED		10,000.00	USDT	client	principal	
CL.AQN.EUR.DUE	8,640.00		EUR	client	principal	client net, in Raeen's sub-account at Aquanow, settled to Raeen's receivers as usual
CO.RAEEN.EUR.SHARE_RECEIVABLE	144.00		EUR	company	share	NPL's remainder: Raeen holds the converted funds and owes it (direction receivable, period)
CO.NONE.EUR.EARN_GROSS	144.00		EUR	company	share	Raeen's share of the net markup; never NPL's money
CO.NONE.EUR.EXP_PARTNER	72.00		EUR	company	partner_cost	Aquanow's stated fee
CL.PROJECT.EUR.PAYABLE		8,640.00	EUR	client	principal	
CO.NONE.EUR.EARN_GROSS		360.00	EUR	company	earnings	gross margin: the sender's 4 %

Share base = fee parts 360.00 – partner cost 72.00 = 288.00 EUR (invariant 25); the rebate below is not in the base. No POOL line: NPL's margin is not resting at the partner but with Raeen, so it is a receivable from Raeen instead. A second party with its own rule on the same leg adds one more pair of lines; the shares may not exceed 100 % of the base.

T33 · Partner rebate accrued with the same conversion (D27): Aquanow returns 0.3 % of the amount = 30.00 USDT source: Rebate

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CO.AQN.USDT.REBATE_RECEIVABLE	30.00		USDT	company	rebate	expected from Aquanow (PARTNER_REBATE_ACCRUAL, state expected, period)
CO.NONE.USDT.EARN_REBATE		30.00	USDT	company	rebate	NPL's alone: never in EARN_GROSS, never visible to Raeen (invariant 26)

The accrual names the NPLify deal it arises on; a partner statement is reconciled only against those deals. A rule with basis pct_of_partner_fee accrues in the currency the partner pays it in (see Section 22).

T34 · Month-end: Aquanow's statement shows 29.50 USDT for the deal and pays it in USDT; the 0.50 USDT difference trues up the income source: Rebate

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CO.OWN.USDT.WALLET	29.50		USDT	company	rebate	received (statement_ref on the accrual, state received)
CO.NONE.USDT.EARN_REBATE	0.50		USDT	company	rebate	true-up against the statement: the accrual's received_amount
CO.AQN.USDT.REBATE_RECEIVABLE		30.00	USDT	company	rebate	the expectation is cleared

A statement line that NPL cannot match to a deal is not posted: NPL only reconciles rebates on NPLify deals (D27).

T35 • Month-end: Raean pays NPL its remainder 144.00 EUR source: MarkupShare

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CO.OWN.EUR.WALLET	144.00		EUR	company	share	received
CO.RAEEN.EUR.SHARE_RECEIVABLE		144.00	EUR	company	share	accrual state settled, received_amount

Had Raean paid in USDT, the receipt would land as Dr CO.OWN.USDT.WALLET / Cr CO.NONE.USDT.FX_CLEARING and the receivable clear as Dr CO.NONE.EUR.FX_CLEARING / Cr CO.RAEEN.EUR.SHARE_RECEIVABLE at the statement's agreed amounts; FX_CLEARING then carries NPL's open currency position (Section 3).

16 • Pattern K — variances, by kind

The sender's figures are write-once, so whenever the partner's actual conversion differs from the one the leg was priced on, the client's entitlement is unchanged and the difference is NPL's. Three of the four variance kinds have the same posting shape and differ only in the account; the fourth, fx timing, belongs to invoices and the own desk (Pattern M).

T36 • Partner calculates (calc_mode partner_calculates): Ali booked 3,000 USDT at 0.8945 instead of the 0.8950 version: expected 2,685.00 EUR, actual 2,683.50 EUR, variance 1.50 EUR

source: Conversion

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.PROJECT.USDT.PAYABLE	3,000.00		USDT	client	principal	
CL.ALI.USDT.COLLECTED		3,000.00	USDT	client	principal	
CL.ALI.EUR.DUE	2,658.00		EUR	client	principal	client net is unchanged: the sender's price is write-once (invariant 5)
CL.ALI.EUR.DUE	0.21		EUR	client	rounding	
CO.ALI.EUR.POOL	25.29		EUR	company	earnings	margin actually left at Ali
CO.NONE.EUR.VAR_CONVERSION	1.50		EUR	company	variance	what the partner did worse than expected; beyond the project's tolerance an exception opens, the deal does not wait
CL.PROJECT.EUR.PAYABLE		2,658.21	EUR	client	principal	
CO.NONE.EUR.EARN_GROSS		26.79	EUR	company	earnings	margin as priced

The same account takes the basis variance of a reroute (the alternate partner's expected conversion on the net amount forwarded).

T37 · Cutoff timing (D29): 3,000 USDT locked at 0.8950 after Ali's 14:30 cutoff, converted next business day at 0.8930: expected 2,685.00 EUR, actual 2,679.00 EUR, variance 6.00 EUR source: Conversion

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.PROJECT.USDT.PAYABLE	3,000.00		USDT	client	principal	
CL.ALI.USDT.COLLECTED		3,000.00	USDT	client	principal	
CL.ALI.EUR.DUE	2,658.00		EUR	client	principal	client net is unchanged: the sender's price is write-once (invariant 5)
CL.ALI.EUR.DUE	0.21		EUR	client	rounding	
CO.ALI.EUR.POOL	20.79		EUR	company	earnings	margin actually left at Ali
CO.NONE.EUR.VAR_CUTOFF	6.00		EUR	company	cutoff_timing	rate stamped at lock vs rate used; NPL absorbs it
CL.PROJECT.EUR.PAYABLE		2,658.21	EUR	client	principal	
CO.NONE.EUR.EARN_GROSS		26.79	EUR	company	earnings	margin as priced

The leg stamps rate_locked_at, cutoff_at and conversion_due_date; Converting may not start before the due date (invariant 28). If the next day's rate would breach the minimum margin, the due date is pushed with conversion_deferred_reason and nothing posts until it runs.

T38 · Honoured expired rate (D45): a new leg priced on yesterday's 0.8950 by an approved override (honour_expired_rate); Ali converts today at 0.8900: expected 2,685.00 EUR, actual 2,670.00 EUR, variance 15.00 EUR source: Conversion

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.PROJECT.USDT.PAYABLE	3,000.00		USDT	client	principal	
CL.ALI.USDT.COLLECTED		3,000.00	USDT	client	principal	
CL.ALI.EUR.DUE	2,658.00		EUR	client	principal	client net is unchanged: the sender's price is write-once (invariant 5)
CL.ALI.EUR.DUE	0.21		EUR	client	rounding	
CO.ALI.EUR.POOL	11.79		EUR	company	earnings	margin actually left at Ali
CO.NONE.EUR.VAR_RATE_HONOUR	15.00		EUR	company	variance	the reused rate vs the day's rate: NPL's cost, chosen (invariant 39)
CL.PROJECT.EUR.PAYABLE		2,658.21	EUR	client	principal	
CO.NONE.EUR.EARN_GROSS		26.79	EUR	company	earnings	margin as priced

The leg stamps rate_reused_from_deal_id, rate_override_reason and rate_override_approved_by (Finance). A variance can also be a gain, in which case the line is a credit.

17 · Pattern L — loss events: booked to NPL first, recovered share by share

Money lost after it was verified or released: a frozen single-use account, cash found short and unrecoverable, a wrong-chain transfer (D32). The whole loss is NPL's expense at once and the client is made whole; the split Management approves is then recovered from each party by its own posting (NPL, 10 October).

T39 • Deal 7 collected: 10,000 USDT verified on a single-use local account at Ali (shown as T1)

source: Collection

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.ALI.USDT.COLLECTED	10,000.00		USDT	client	principal	
CL.PROJECT.USDT.PAYABLE		10,000.00	USDT	client	principal	

T40 • Loss event (D32): the account is frozen before conversion and the 10,000.00 USDT are lost; NPL books the whole loss and makes the client whole

source: LossEvent

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CO.NONE.USDT.EXP_LOSS	10,000.00		USDT	company	loss	the whole loss is NPL's first (LOSS_EVENT state approved → posted)
CO.ALI.USDT.POOL	6,000.00		USDT	company	loss	NPL's money at Ali funds the make-good as far as it goes
CO.OWN.USDT.WALLET	4,000.00		USDT	company	loss	the rest is transferred from NPL's own wallet to Ali (partner_agreement_ref)
CL.ALI.USDT.COLLECTED		10,000.00	USDT	client	principal	the client money that was lost
CL.ALI.USDT.COLLECTED	10,000.00		USDT	client	loss	replaced by NPL's money: the deal proceeds, the collection part stays verified

The client's claim (PAYABLE) is untouched and the deal converts and settles normally. When neither the pool nor an immediate transfer can fund the make-good, the unfunded part is posted as Dr CL.PROJECT.USDT.MAKEGOOD / Cr CO.NONE.USDT.LOSS_PAYABLE instead and the client balance at Ali is restored only when NPL's funds land there (then Dr CL.ALI.USDT.COLLECTED / Cr CL.PROJECT.USDT.MAKEGOOD and Dr CO.NONE.USDT.LOSS_PAYABLE / Cr CO.OWN.USDT.WALLET). A loss on the settlement side credits INTRANSIT instead of COLLECTED and restores DUE, from which a new settlement follows. A collection is moved to state reversed only when Management decides the deal is unwound instead; the make-good then becomes a refund to the sender.

T41 • Management approves the split (approve_loss_split): partner 4,000.00, sender 2,000.00, client 1,000.00, NPL 3,000.00 USDT source: LossEvent

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CO.ALI.USDT.LOSS_RECEIVABLE	4,000.00		USDT	company	loss	Ali's agreed share, now owed to NPL
CO.SENDER_A.USDT.LOSS_RECEIVABLE	2,000.00		USDT	company	loss	the sender's agreed share
CO.ALI.USDT.POOL	1,000.00		USDT	company	loss	NPL takes back the part of its make-good the client bears
CO.NONE.USDT.EXP_LOSS		7,000.00	USDT	company	loss	NPL's expense falls to its own share, 3,000.00
CL.PROJECT.USDT.PAYABLE	1,000.00		USDT	client	loss	the client's claim falls by the share it bears
CL.ALI.USDT.COLLECTED		1,000.00	USDT	client	loss	and the restored balance with it

Each share is a separate recovery posting against the loss; the shares sum to the amount (invariant 31). The client-share lines are the partial reversal of the make-good in T40, which is why invariant 16 is restated in Section 22.

T42 • Ali settles its share: 4,000.00 USDT credited to NPL's pool at Ali source: LossEvent

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CO.ALI.USDT.POOL	4,000.00		USDT	company	loss	received (LOSS_RECOVERY, Section 22)
CO.ALI.USDT.LOSS_RECEIVABLE		4,000.00	USDT	company	loss	

T43 • The sender settles its share: 2,000.00 USDT paid to NPL's own wallet source: LossEvent

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CO.OWN.USDT.WALLET	2,000.00		USDT	company	loss	a payment to NPL, never a netting against a sender credit (client money)
CO.SENDER_A.USDT.LOSS_RECEIVABLE		2,000.00	USDT	company	loss	

Loss accounts after the recoveries

Account	Ccy	Balance	Meaning
CO.NONE.USDT.EXP_LOSS	USDT	3,000.00	NPL's own share of the loss, once every recovery is settled
CO.ALI.USDT.LOSS_RECEIVABLE	USDT	0.00	nothing outstanding from Ali
CO.SENDER_A.USDT.LOSS_RECEIVABLE	USDT	0.00	nothing outstanding from the sender

Net effect on NPL's money: the pool at Ali gave 6,000 and received 1,000 + 4,000 back (−1,000); the own wallet gave 4,000 and received 2,000 (−2,000); together −3,000, NPL's share. The client's claim on deal 7 is 9,000 USDT, backed by 9,000 at Ali.

18 · Pattern M — the reseller project (NPL-GR): full reseller P&L

NPL-GR, a business unit of NPL, buys from vendors against invoices and sells to reseller customers at a retail price; the own desk (OWN_WALLET role own_desk, LT Sub at Aquanow; PARTNER_CONFIG.is_own_desk) holds and converts the funds (D23, D24). NPL's decision of 10 October: run the full reseller P&L inside NPLify. The money is NPL's from the moment it arrives, so every line is company-tagged; the customer's price is revenue, the vendor's invoice is cost, and the margin is read in one currency per deal. The example uses NPL's own figures: a 9,000 EUR invoice paid in USDT in two instalments at 0.8708 and 0.87527.

T44 · NPL-GR (reseller): the vendor's invoice of 9,000.00 EUR is recorded (INVOICE, open); the customer is quoted 10,547.00 USDT (invoice ÷ 0.8708, + 2 % markup, rounded up) and pays it

source: Collection

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CO.OWNDESK.USDT.COLLECTED	10,547.00		USDT	company	principal	NPL's money from the moment it arrives: the project's client is NPL-GR itself
CO.NONE.USDT.RESELL_REVENUE		10,547.00	USDT	company	resale	the retail price, recognised when collected

Recording the invoice posts nothing: the obligation is INVOICE_BALANCE (amount – Σ obligation_discharged), and the cost is booked as it is discharged, in the currency actually paid, so that revenue, cost and margin of a deal are in one currency.

T45 · The own desk makes the funds available (a same-currency conversion at rate 1, D23, D33): 10,547.00 USDT ready to pay the vendor source: Conversion

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CO.OWNDESK.USDT.DUE	10,547.00		USDT	company	principal	in NPL's own wallet (LT Sub at Aquanow), reported by the custody view
CO.OWNDESK.USDT.COLLECTED		10,547.00	USDT	company	principal	

No EARN_GROSS in a reseller project: the whole margin is revenue minus cost, so the conversion carries no fee parts. EXP_PARTNER is zero by construction (the own desk is NPL).

T46 · Instalment 1: 3,445.11 USDT paid to the vendor, discharging 3,000.00 EUR of the invoice at discharge_rate 0.8708 (SETTLEMENT_LINE.invoice_id, obligation_discharged) source:

Settlement

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CO.OWNDESK.USDT.INTRANSIT	3,445.11		USDT	company	principal	
CO.OWNDESK.USDT.DUE		3,445.11	USDT	company	principal	

T47 • The vendor confirms instalment 1: cost of 3,000.00 EUR recognised at the priced rate = 3,445.11 USDT; no timing difference on the quote day source: Confirmation

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CO.NONE.USDT.RESELL_COST	3,445.11		USDT	company	resale	the discharged part of the vendor's invoice, at the rate the customer's price was built on
CO.OWNDESK.USDT.INTRANSIT		3,445.11	USDT	company	principal	delivered; INVOICE_BALANCE falls to 6,000 EUR

T48 • Instalment 2, four days later: 6,855.03 USDT discharge the remaining 6,000.00 EUR at 0.87527; the vendor confirms (settlement and confirmation shown together) source: Confirmation

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CO.OWNDESK.USDT.INTRANSIT	6,855.03		USDT	company	principal	released
CO.OWNDESK.USDT.DUE		6,855.03	USDT	company	principal	
CO.NONE.USDT.RESELL_COST	6,890.22		USDT	company	resale	the remaining 6,000 EUR at the priced rate
CO.NONE.USDT.VAR_FX_TIMING		35.19	USDT	company	fx_timing	the rate moved in NPL's favour between pricing and payment (a debit when it moves against)
CO.OWNDESK.USDT.INTRANSIT		6,855.03	USDT	company	principal	delivered; INVOICE_BALANCE is zero, INVOICE settled

Reseller margin on the deal = revenue 10,547.00 – cost 10,335.33 + fx timing 35.19 = 246.86 USDT, which is exactly what is left in CO.OWNDESK.USDT.DUE. NPL accepts the forex difference within the 2 % markup (D24); the account shows how much of the markup it consumed.

Reseller balances after the invoice is settled

Account	Ccy	Balance	Meaning
CO.OWNDESK.USDT.COLLECTED	USDT	0.00	nothing unconverted
CO.OWNDESK.USDT.DUE	USDT	246.86	NPL's margin on the deal, resting in the own desk (custody view)
CO.OWNDESK.USDT.INTRANSIT	USDT	0.00	nothing in transit
CO.NONE.USDT.RESELL_REVENUE	USDT	10,547.00	retail price
CO.NONE.USDT.RESELL_COST	USDT	10,335.33	vendor invoice at the priced rate
CO.NONE.USDT.VAR_FX_TIMING	USDT	-35.19	rate movement between pricing and payment (negative = gain)

T49 • Variant: the own desk converts 10,547.00 USDT to EUR to pay the vendor in EUR; expected 9,492.30 EUR at 0.9 (the rate the price was built on), obtained 9,458.55 at 0.8968 source:

Conversion

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CO.NONE.USDT.RESELL_REVENUE	10,547.00		USDT	company	resale	the revenue leaves the collected currency
CO.OWNDESK.USDT.COLLECTED		10,547.00	USDT	company	principal	
CO.OWNDESK.EUR.DUE	9,458.55		EUR	company	principal	what the desk obtained
CO.NONE.EUR.VAR_FX_TIMING	33.75		EUR	company	fx_timing	the price was fixed in USDT; the rate moved before NPL converted
CO.NONE.EUR.RESELL_REVENUE		9,492.30	EUR	company	resale	the revenue re-stated in the vendor's currency at the priced rate

Revenue is re-denominated the way a client obligation is in T2, so the margin is then read in EUR: 9,492.30 – 9,000 (cost booked in EUR as the instalments are confirmed) – 33.75 fx timing = 458.55 EUR. This variant is shown for the posting shape; the balances above are the USDT path only.

Bank fees on a vendor payment, network fees and a bounced vendor transfer post exactly as in Patterns C and F, with owner company on every line. A loss event in the reseller project has no make-good: the loss is NPL's and the recoveries are the same as Pattern L without the client-share lines.

19 · Identities and controls

Identity	Statement
Client assets = client liabilities, per currency	$\Sigma \text{ COLLECTED} + \Sigma \text{ HELD} + \Sigma \text{ DUE} + \Sigma \text{ INTRANSIT} + \Sigma \text{ SHORTFALL} + \text{MAKEGOOD} = \text{PAYABLE} + \Sigma \text{ CREDIT}$. Holds after every transaction because every transaction balances per owner as well as per currency.
PAYABLE = Σ GROUP_ENTITLEMENT	Per project and currency (invariant 7). Every payout line resolves to one group — the paid receiver's for a counterparty, the paying leg's via deal_id otherwise — so the identity is checked line by line (invariant 19).
PROJECT_BALANCE(partner) = CL.<PARTNER>.<CCY>.DUE	The view and the account agree; the pending converted value of a locked, not yet converted leg (D29) is shown by the view and is not posted.
Pool per partner = the partner's statement	$\text{CO}.\text{<PARTNER>}.<\text{CCY}>.\text{POOL}$ agrees to what the partner says it holds for NPL; $\Sigma \text{ EARNINGS_RECEIVABLE}$ outstanding $\leq \text{POOL}$.
Rebates and shares agree to statements	$\Sigma \text{ REBATE_RECEIVABLE} = \Sigma \text{ PARTNER_REBATE_ACCRUAL}$ expected – received, per partner; $\Sigma \text{ SHARE_RECEIVABLE} / \text{SHARE_PAYABLE} = \Sigma \text{ MARKUP_SHARE_ACCRUAL}$ not yet settled, per party.
Loss accounting closes	Per LOSS_EVENT : EXP_LOSS net of recoveries = npl_share once every share is settled; $\Sigma \text{ LOSS_RECEIVABLE} = \text{agreed shares not yet received}$.
Exposure $\leq$ ceiling	$\Sigma \text{ DUE} + \Sigma \text{ INTRANSIT}$ at a partner is within the project's exposure ceiling; $\Sigma \text{ PROJECT_BALANCE}$ across a client's projects, as REPORTING_VALUE , within $\text{CLIENT.exposure_ceiling}$ (invariant 35).
Own wallets hold client money only in three cases	$\text{CL}.\text{OWN}.\text{<CCY>}. \text{HELD}$ is zero except for rows the custody view can name: an open reroute, a bridge collection within custody_max_hours, an own-desk leg (invariant 3).
Each confirmation = its released amount	Confirmed + shortfall + fees booked = the amount released on the line.
Reseller P&L closes	In a reseller project: $\text{RESELL_REVENUE} - \text{RESELL_COST} \pm \text{VAR_FX_TIMING} - \text{EXP_*}$ on the deal = what remains in $\text{CO}.\text{OWNDESK}.*.\text{DUE}$ after the invoice is settled, per currency.
Every figure shown is a sum	No screen stores a balance; all are sums over postings or deals (invariant 10). A converted total is a REPORTING_VALUE : an estimate at the latest platform-pair rate, never posted (invariant 38).

Month-end, Finance runs the identities above and reconciles three statements: each partner's statement of what it holds for NPL (POOL), each partner's rebate statement (REBATE_RECEIVABLE , NPLify deals only), and each markup-share party's statement ($\text{SHARE_RECEIVABLE} / \text{SHARE_PAYABLE}$). A difference is a true-up posting that names the statement line, never an edit.

20 · Reports the ledger answers

- **Where the money is.** Client money by partner, in transit, in custody (CUSTODY view: reroutes, bridge, own desk), shortfalls and credits; NPL's money by partner (POOL) and wallet (WALLET), receivables and payables by counterparty.
- **Client position.** PROJECT_BALANCE per partner and GROUP_ENTITLEMENT per group, rolled up per client (CLIENT_POSITION) with the period's EARN_GROSS and EARN_REBATE ; compared with the client's exposure ceiling as a REPORTING_VALUE .

- **Profit and loss, per currency.** EARN_GROSS (net of shares) + EARN_REBATE + RESELL_REVENUE – RESELL_COST – EXP_PARTNER – EXP_BANKFEE – EXP_NETWORK – EXP_ROUNDING – EXP_LOSS ± the four VAR_ accounts, by project, partner, pair and month, from the cost-component and source tags. A total across currencies is a REPORTING_VALUE with its rate and time.
- **Partner commissions.** Per partner and period: margin captured there, dues outstanding (EARNINGS_RECEIVABLE), rebates expected, reconciled and received (statement_ref).
- **Markup shares.** Per party and period: base, share, direction, settled and open amounts; a party never sees a rebate.
- **Variance and loss.** The four variance kinds by partner and month, each line traceable to its leg; loss events with their shares and recovery state.
- **Reseller margin.** Per NPL-GR deal: revenue, cost, fx timing, fees, margin, in the currency of the deal; INVOICE_BALANCE per vendor invoice.

21 · Transaction sources and what they post

Source event	Posts to	Owner tags	Cost components
Collection	COLLECTED or HELD, PAYABLE, CREDIT; in a reseller project COLLECTED and RESELL_REVENUE	client (reseller: company)	principal, resale
Conversion (deal leg)	COLLECTED / PAYABLE in the in-currency; DUE, PAYABLE, POOL, EARN_GROSS, EXP_PARTNER, EXP_ROUNDING, SHARE_RECEIVABLE, a VAR_ account in the out-currency; reseller: COLLECTED → DUE, RESELL_REVENUE re-denominated, VAR_FX_TIMING	client and company	principal, rounding, earnings, partner_cost, share, variance, cutoff_timing, fx_timing
BalanceConversion	DUE and PAYABLE in both currencies; POOL and EARN_GROSS if a markup is decided	client (and company)	principal, rounding, earnings
Settlement	INTRANSIT, DUE; EXP_BANKFEE and POOL when a fee is absorbed up front; EXP_NETWORK and WALLET on a hop. Each line names the leg it settles (deal_id) and, in NPL-GR, the invoice it discharges	client (and company)	principal, bank_fee, network_fee
SettlementReturn	DUE, INTRANSIT; EXP_BANKFEE and POOL for the bounce fee	client and company	principal, bank_fee
Confirmation	PAYABLE, INTRANSIT, SHORTFALL; a hop: COLLECTED at the partner; reseller: RESELL_COST, VAR_FX_TIMING	client (reseller: company)	principal, bank_fee, resale, fx_timing
BankFeeEvent	EXP_BANKFEE, POOL, DUE, SHORTFALL	company and client	bank_fee
Reroute	HELD, COLLECTED at the alternate partner; WALLET and POOL for recovered dues; VAR_CONVERSION for basis variance	client and company	principal, earnings, variance
MarkupShare	EARN_GROSS, SHARE_PAYABLE or SHARE_RECEIVABLE; WALLET or POOL on settlement	company	share

Source event	Posts to	Owner tags	Cost components
Rebate	REBATE_RECEIVABLE, EARN_REBATE; WALLET on receipt	company	rebate
LossEvent	EXP_LOSS, POOL or WALLET (or LOSS_PAYABLE), COLLECTED / INTRANSIT / DUE (or MAKEGOOD); then LOSS_RECEIVABLE, PAYABLE for the recoveries	company and client	loss, principal
Adjustment	any, always as a reversal plus a correct re-posting, a Management-approved disposition, or a cross-currency clearing through FX_CLEARING	as reversed	as reversed

22 · Who sees what, and who may post

Rights are rows, not text (D44): `ROLE` holds the three roles of today, operations, finance and management, and `ROLE_RIGHT` says per controlled action which role may initiate and which may approve, with the role added on escalation; the approver is never the initiator (invariant 9).

- **Operations** (sees_economics false) sees operational fields, sender-facing prices and partner rates as entered, but never a ledger balance, a margin, a pool, a share, a rebate, a cost breakdown or a variance, on any screen, export, message or report.
- **Finance** sees all postings and balances, owns reconciliation and the month-end true-ups, initiates adjustments, balance conversions and dues recovery, recommends covered bank fees and fee caps (`FEE_DECISION.recommended_by`), approves honoured expired rates and quote-validity extensions, and overrides a screening reject.
- **Management** approves the actions the ledger patterns depend on: `approve_loss_split`, `balance_conversion`, `recover_dues`, `cross_group_payout`, `cover_bank_fee`, `raise_fee_cap`, `enable_own_desk`, settlements above limit and past the cutoff, own-wallet outbound transfers, and post-money voids. Every controlled action is audited with actor, time, reason and record version.

23 · What this design asks of the ERD (proposed D46)

The patterns above need the following additions to Draft v5.6. Each is small; none changes an existing posting. They are listed for NPL's agreement and will be cut into the next ERD edition together.

1. **Variance by kind.** `LEDGER_ACCOUNT.purpose`: `VARIANCE` replaced by `VAR_CONVERSION`, `VAR_CUTOFF`, `VAR_RATE_HONOUR`, `VAR_FX_TIMING`. Invariants 28 and 39 and decisions D24, D29, D45 then name the account they post to.
2. **Loss first, recover later.** Purposes `LOSS_RECEIVABLE` (holder Partner or Sender), and the pair `MAKEGOOD` (client asset, holder Project) / `LOSS_PAYABLE` (company liability) for an unfunded make-good. A table `LOSS_RECOVERY` (`loss_event_id`, party kind sender / partner / client, amount, state agreed / received, `received_at`, reference, `approved_by`) as the source record of each recovery posting, under source `LossEvent`. `COLLECTION.state reversed` is used only when the deal is unwound. Invariant 16 restated: ownership flips client → company at margin recognition and, by partial reversal of a make-good, when a Management-approved loss share is borne by the client.
3. **Accrual timing.** `MARKUP_SHARE_ACCRUAL` and `PARTNER_REBATE_ACCRUAL` are created and posted by the conversion that gives rise to them (invariant 25 and 26 restated to say so); `PARTNER_REBATE_RULE.payout_currency` names the currency the partner pays in, in which the accrual is expressed (the amount basis for `pct_of_amount`, the conversion's own rate for `pct_of_partner_fee`).

4. **Reseller project.** PROJECT.ownership_model client_money / own_business (own_business requires is_own_desk on its partner configuration and Management approval, like enable_own_desk); purposes RESELL_REVENUE, RESELL_COST and the where-is-it purposes under owner CO; cost component resale; SETTLEMENT_LINE.obligation_discharged and discharge_rate drive the cost posting; a view RESELLER_MARGIN(deal_group) = revenue – cost ± fx timing – fees. D23's note that own-desk deals use the client-money accounts applies only to an own desk inside a client_money project.
5. **Cross-currency company settlements.** Purpose FX_CLEARING (holder None), used only by Finance adjustments when a rebate, share or loss share is paid in a currency other than the one it accrued in; its balances are reported as NPL's open currency position through REPORTING_VALUE.
6. **Holder types.** LEDGER_ACCOUNT.holder_type gains Client (reserved) and keeps Sender for LOSS_RECEIVABLE; OwnDesk is the own-desk PARTNER_CONFIG, not a new type.

Draft v1.2 — for review with NPL. Figures are worked examples, not NPL data, except the invoice instalment rates of Pattern M, which are NPL's.